EXPERIMENT 14 – Pro-Code Custom Functions

Project: Student Academic Performance Tracker using Low-Code in Zoho Creator

AIM

To design a custom function for validating student academic marks and calculating the total marks, average, and academic performance.

REQUIREMENTS

- Student form with student name, register number, and subject marks.
- A custom function or workflow action supported by the Zoho Creator account.
- Fields to store total marks, average, and performance status.

DESIGN / MODULES

Input / Output	Validation / Use
Student Name	Must be present
Register Number	Must be present
Subject Marks	Must be between 0 and 100
Total Marks	Sum of all subject marks
Average	Total marks divided by number of subjects
Performance Status	Grade based on average

PROCEDURE

1. Create a Student Academic Performance form in Zoho Creator.
2. Add fields for student name, register number, and marks in three subjects.
3. Create fields for total marks, average, and performance status.
4. Create a custom function or workflow action using the Deluge editor supported by the account.
5. Validate the student details and ensure all marks are between 0 and 100.
6. Calculate total marks and average, then assign a performance status.
7. Store the calculated values using the correct field link names.
8. Test valid marks, missing details, and invalid marks before deploying the workflow.

SAMPLE LOGIC / OUTPUT

Illustrative Deluge logic only. Adjust field link names to match your Zoho Creator form.

```
if(input.Subject1 != null && input.Subject2 != null && input.Subject3 != null)
{
    if(input.Subject1 >= 0 && input.Subject1 <= 100 &&
       input.Subject2 >= 0 && input.Subject2 <= 100 &&
       input.Subject3 >= 0 && input.Subject3 <= 100)
    {
        total = input.Subject1 + input.Subject2 + input.Subject3;
        average = total / 3.0;

        if(average >= 90)
        {
            status = "Excellent";
        }
        else if(average >= 75)
        {
            status = "Good";
        }
        else if(average >= 50)
        {
            status = "Average";
        }
        else
        {
            status = "Needs Improvement";
        }

        info total;
```

```
        info average;
        info status;
    }
    else
    {
        info "Marks must be between 0 and 100.";
    }
}
else
{
    info "Please enter marks for all subjects.";
}
```

TEST CASES

Input	Expected Outcome
Marks: 90, 80, 70	Total: 240, Average: 80, Status: Good
Marks: 100, 95, 95	Total: 290, Average: 96.67, Status: Excellent
Marks: 40, 50, 45	Total: 135, Average: 45, Status: Needs Improvement
Marks: 110, 80, 70	Validation message; no calculation
Missing subject mark	Validation message

RESULT

The custom-function logic and test cases for the Student Academic Performance Tracker were documented. The function validates subject marks, calculates total marks and average, and determines the student's performance status. The illustrative code must be adapted to the actual Zoho Creator field link names and Deluge syntax before deployment.